DRUGS · EAR PREPARATIONS

Acidifying and astringent ear preparations

Healthy ear canal skin is acidic, with a pH of about 5 to 6, which keeps Pseudomonas and other bacteria in check. Acidifying preparations (acetic acid), astringents (aluminium acetate) and drying powders (boric acid) restore that environment. They are useful for mild otitis externa, for prevention in swimmers and in wet mastoid cavities, and as an alternative to antibiotics.

1 CHAPTER ONE At a glance

The preparations on one page.

1.1 Summary

ACIDIFYING AND ASTRINGENT EAR PREPARATIONS

	SUMMARY
Preparations	Acetic acid 2% spray or drops (alone or with a steroid); aluminium acetate 8% to 13% solution (Burow's solution), often on a wick; boric acid powder (insufflation) or solution; spirit (alcohol) drops for prevention
How they work	Lower the canal pH, are bacteriostatic and antifungal, and dry and shrink inflamed skin
ENT uses	Mild acute otitis externa; prevention in swimmers and recurrent otitis externa; wet, discharging mastoid cavities; adjunct in otomycosis
Dose	Acetic acid spray, one spray three times daily for up to 7 days (until 2 days after symptoms resolve)
Avoid	Perforated drum for acetic acid and alcohol drops (pain and possible ototoxicity)

2 CHAPTER TWO How they work

Mechanism and handling.

2.1 Mechanism

Acid inhibits Pseudomonas and staphylococci, astringents precipitate proteins and reduce oedema and exudate, and boric acid both dries and acidifies.

3 CHAPTER THREE Using them in ENT

Indications and doses.

3.1 Mild otitis externa and prevention

Acetic acid spray is an over-the-counter first step in mild otitis externa, with results close to antibiotic-steroid drops in the first week but less effective with longer courses. Swimmers with recurrent infection can use acidifying or alcohol drops after swimming.

3.2 Cavities

Insufflating **boric acid powder** after cleaning a wet mastoid cavity helps keep it dry; repeated sessions may be needed.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

Stinging and burning (especially in an inflamed canal), and irritation through a perforation.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The normal pH of the canal and why acidification helps.
- Where acidifiers fit in otitis externa and in mastoid cavity care.

5.2 Common pitfalls

COMMON PITFALLS

- Acetic acid or alcohol drops in an ear with a perforation.

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